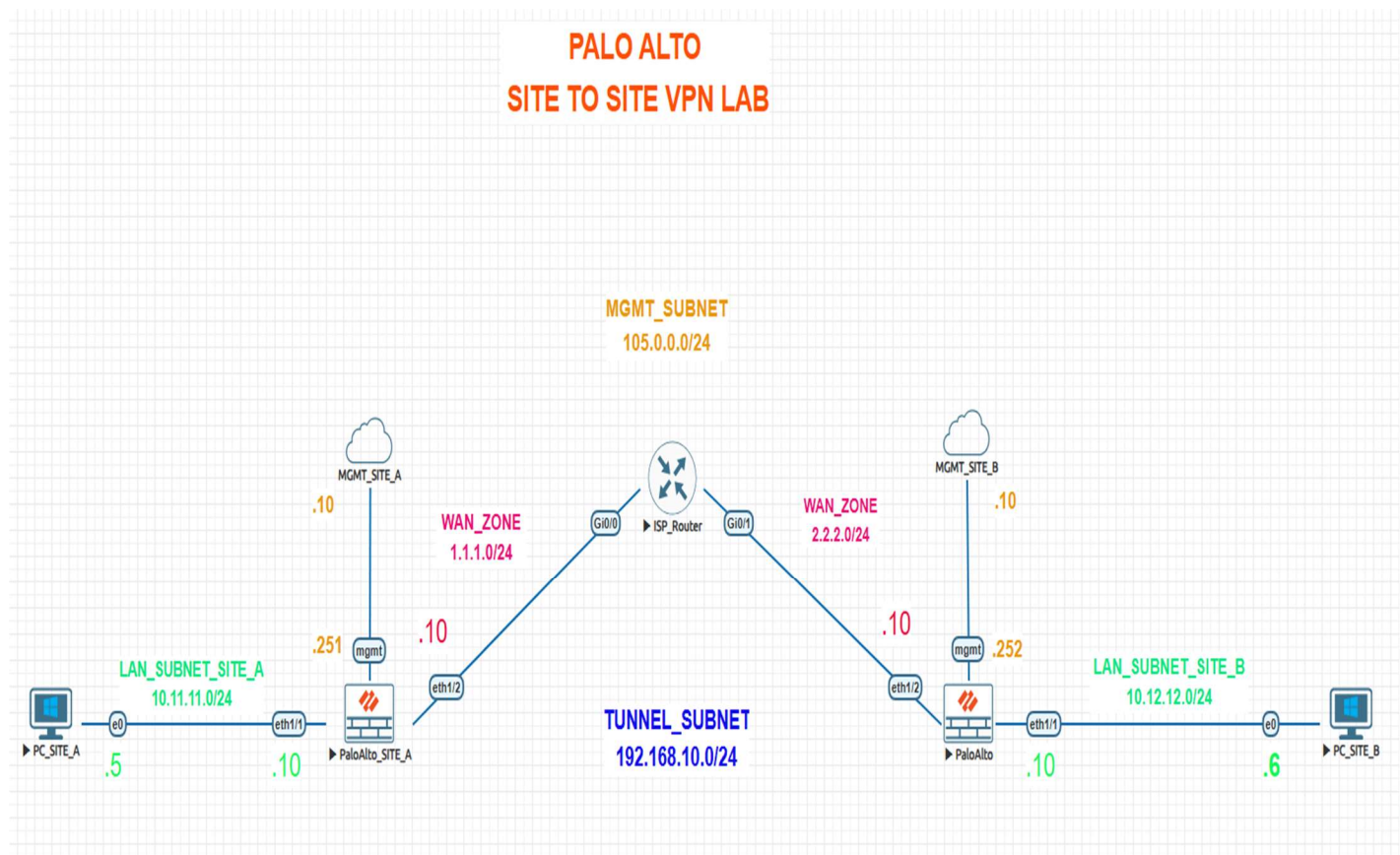

PALO ALTO FIREWALL SITE TO SITE VPN LAB

TOPOLOGY



Network topology of Site-to-Site VPN between Palo Alto Firewalls

Detailed Description (for documentation):

This diagram should display two Palo Alto Firewalls (Site A and Site B) connected through the Internet (Public Network). Each firewall should connect to a LAN network with one PC.

- Site A Firewall WAN IP: **105.0.0.251**
- Site B Firewall WAN IP: **105.0.0.252**
- Site A LAN: **192.168.1.0/24**
- Site B LAN: **192.168.2.0/24**
- Label tunnel interface tunnel.1 between both sites.
Optionally, show “IKE Phase 1” and “IPSec Phase 2” across the WAN link.

INITIAL CONFIGURATIONS

=====

PaloAlto_SITE_A - Setup Guide

=====

STEP 1: Login with Default Credentials

login: admin (Default username)

password: admin (Default password)

STEP 2: Change Admin Password

Enter old password: admin (Current password)

Enter new password: Ab@12345 (New strong password)

Confirm password: Ab@12345 (Confirm new password)

STEP 3: Enter Configuration Mode

> configure (Enter configuration mode)

STEP 4: Remove DHCP Client Mode

delete deviceconfig system type dhcp-client (Remove DHCP if previously enabled)

STEP 5: Set Static IP Address

set deviceconfig system ip-address 105.0.0.251 netmask 255.255.255.0 (Assign static IP and subnet mask)

STEP 6: Commit Changes

commit (Apply and save config)

STEP 7: Exit Configuration Mode

exit (Return to operational prompt)

STEP 8: Verify Interface Settings

> show interface management (View current IP, MAC, status, etc.)

=====

After setup, access GUI at:

https://105.0.0.251

=====

=====

PaloAlto_SITE_B - Setup Guide

=====

STEP 1: Login with Default Credentials

login: admin (Default username)
password: admin (Default password)

STEP 2: Change Admin Password

Enter old password: admin (Current password)
Enter new password: Ab@12345 (New strong password)
Confirm password: Ab@12345 (Confirm new password)

STEP 3: Enter Configuration Mode

> configure (Enter configuration mode)

STEP 4: Remove DHCP Client Mode

delete deviceconfig system type dhcp-client (Remove DHCP if previously enabled)

STEP 5: Set Static IP Address

set deviceconfig system ip-address 105.0.0.252 netmask 255.255.255.0 (Assign static IP and subnet mask)

STEP 6: Commit Changes

commit (Apply and save config)

STEP 7: Exit Configuration Mode

exit (Return to operational prompt)

STEP 8: Verify Interface Settings

> show interface management
(View current IP, MAC, status, etc.)

=====

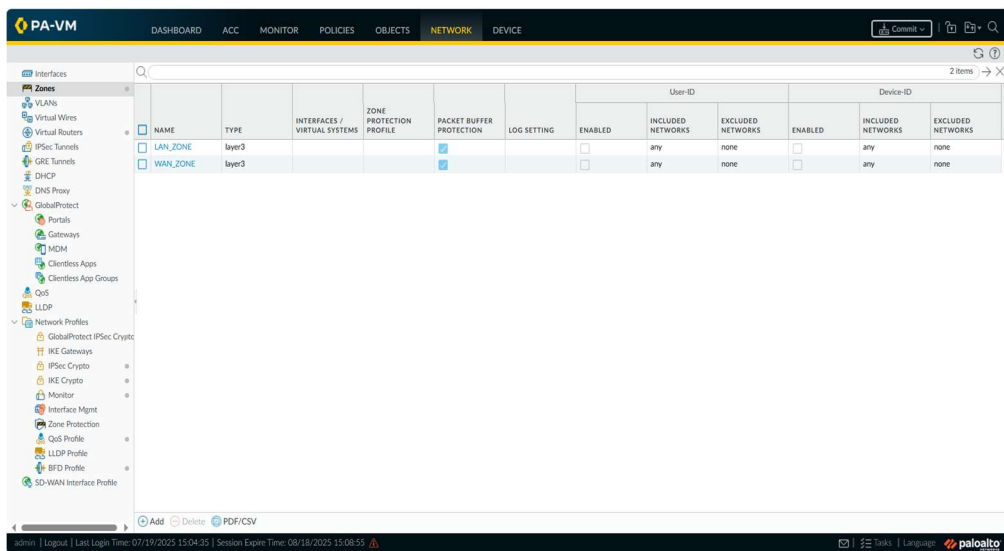
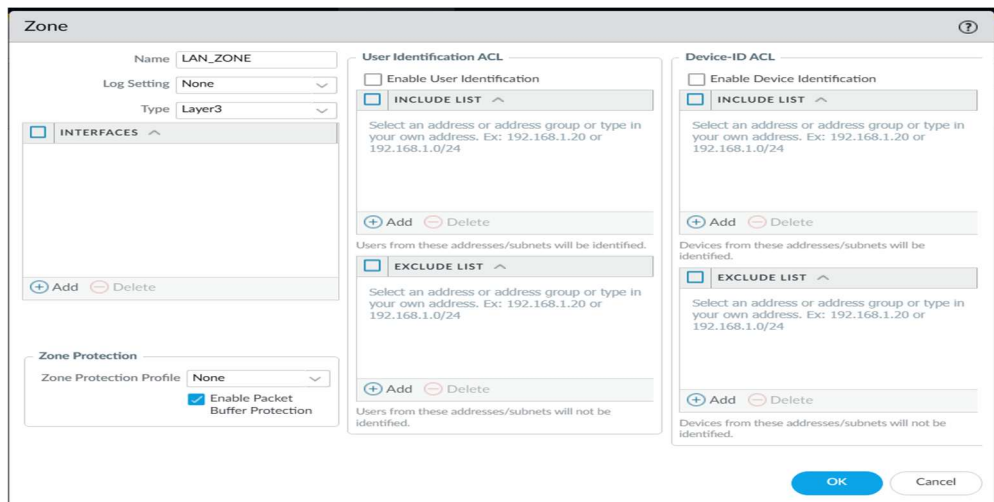
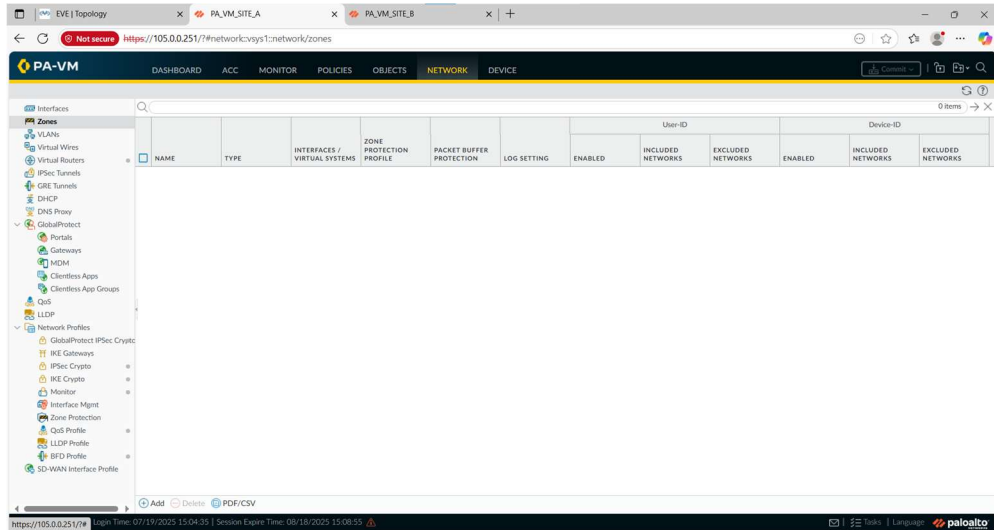
After setup, access GUI at:

https://105.0.0.252

=====

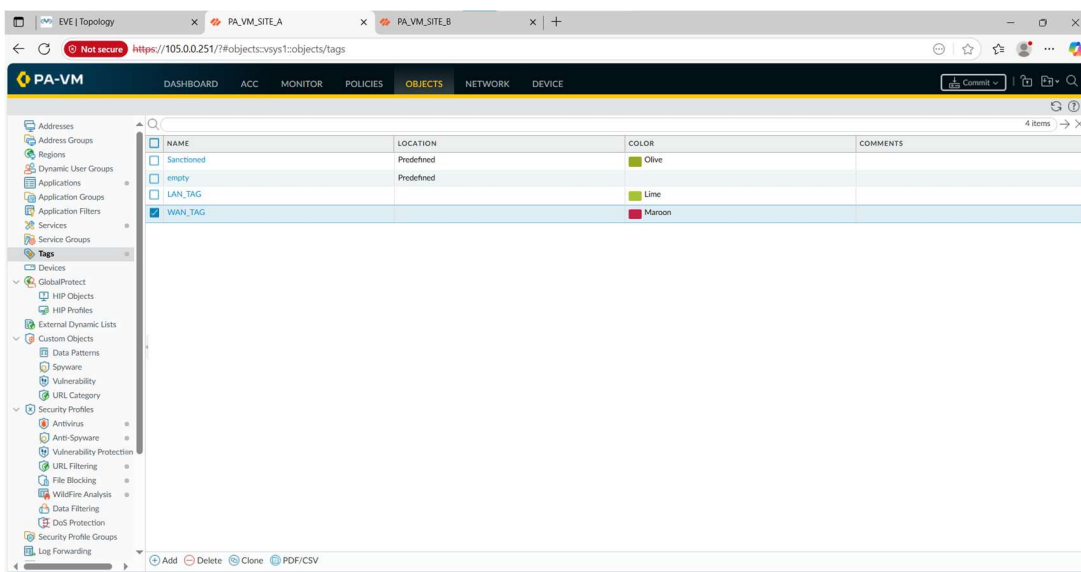
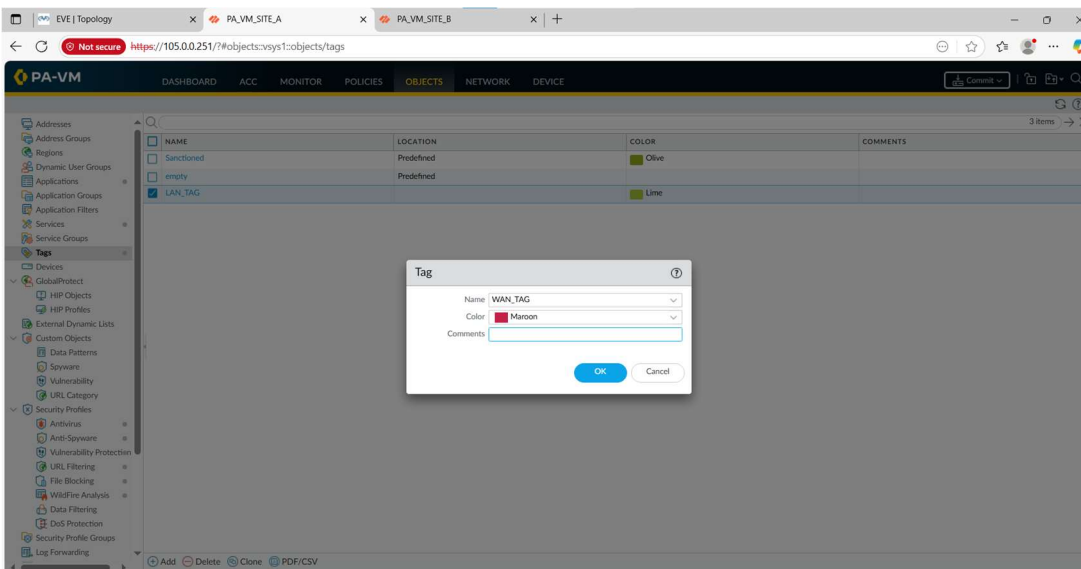
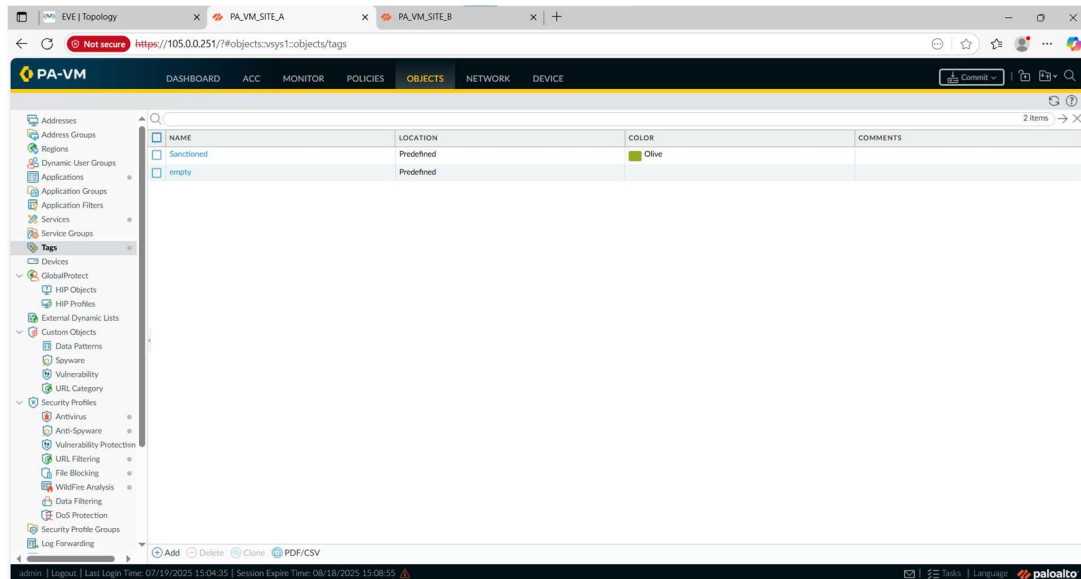
1. To Create Zone

Go to network > zone > add



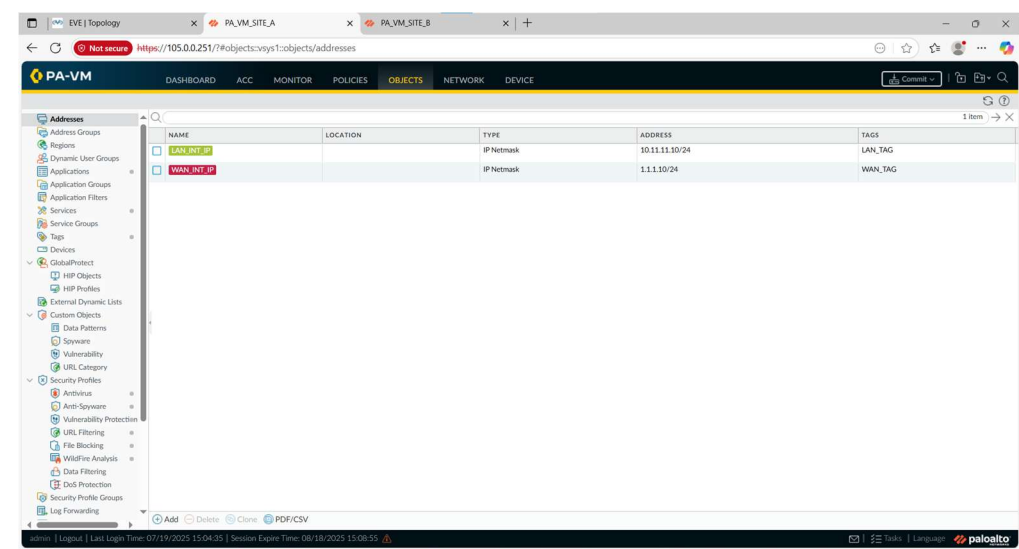
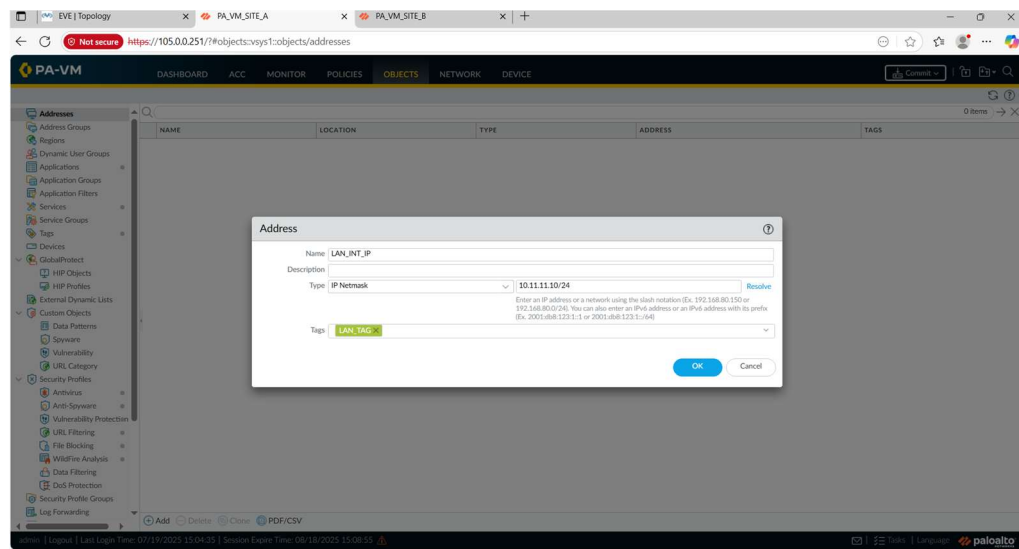
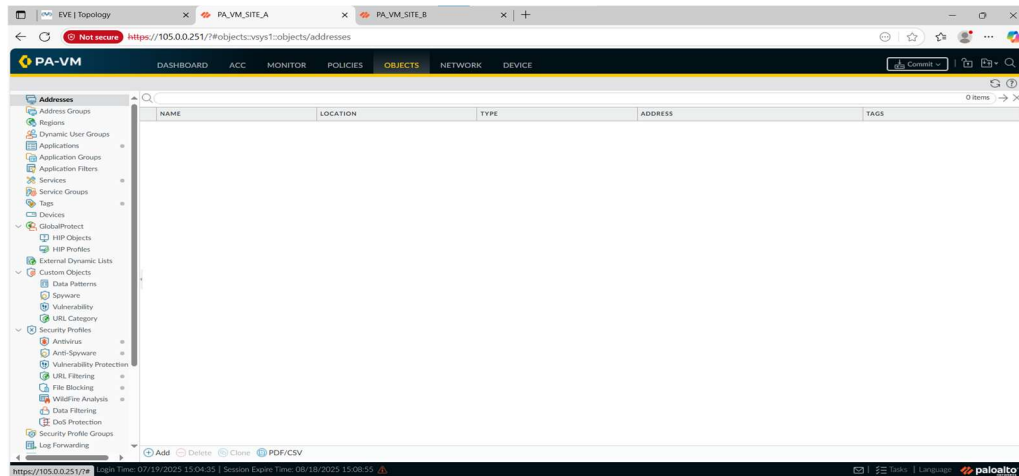
2. To Create Tags

Go to objects>tag>add



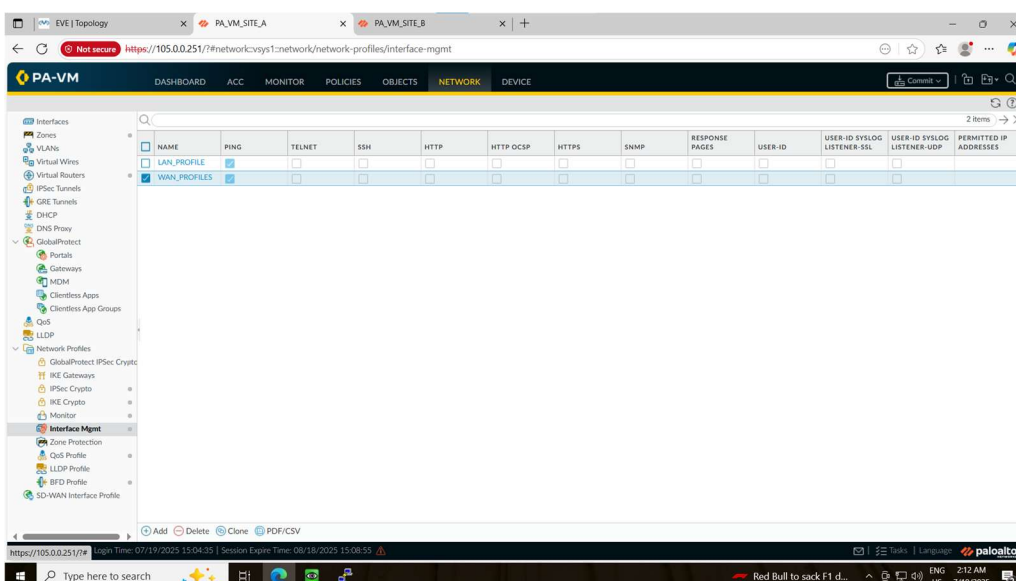
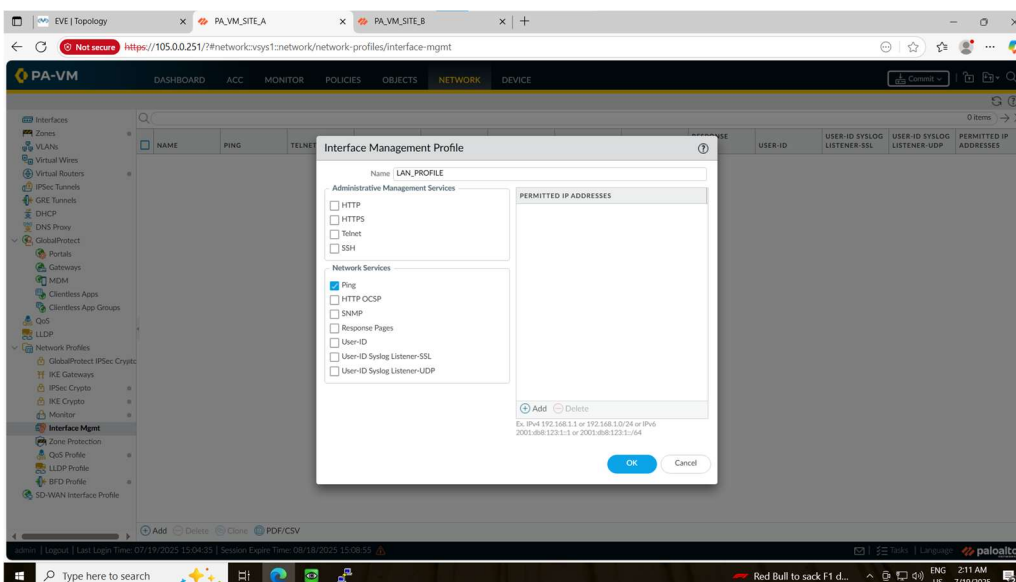
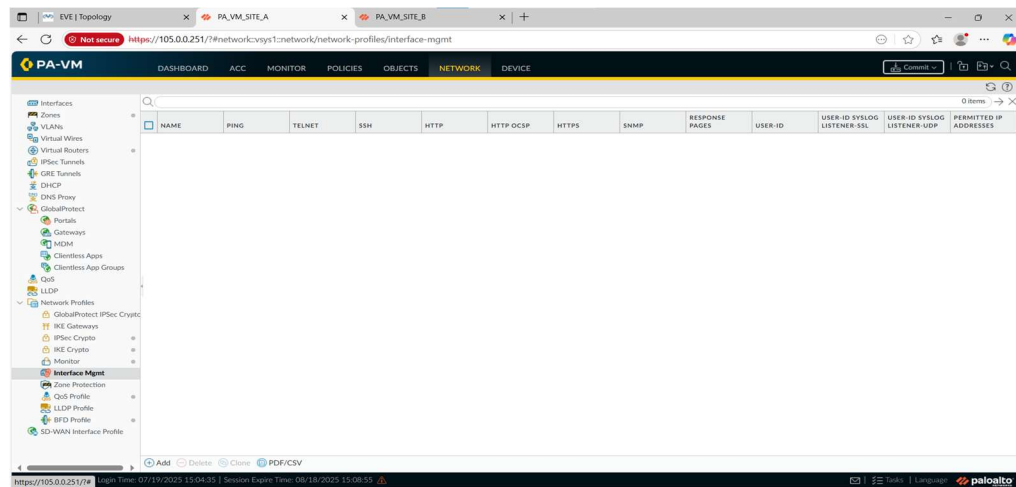
3. To create objects for ip address

Objects >Addresses>Add



4. To create interface management profile (to enable particular services for given interface)

Network > network profile > interface Mgmt



5. Assigning all above configuration to interface

Network>interfaces>select interface

PA-VM

Interfaces

Ethernet | VLAN | Loopback | Tunnel | SD-WAN

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE	SD-WAN INTERFACE PROFILE	FEATURES	COMMENT
ethernet1/1				none	none	Untagged	none	none			
ethernet1/2				none	none	Untagged	none	none			
ethernet1/3				none	none	Untagged	none	none			
ethernet1/4				none	none	Untagged	none	none			
ethernet1/5				none	none	Untagged	none	none			
ethernet1/6				none	none	Untagged	none	none			
ethernet1/7				none	none	Untagged	none	none			
ethernet1/8				none	none	Untagged	none	none			
ethernet1/9				none	none	Untagged	none	none			
ethernet1/10				none	none	Untagged	none	none			
ethernet1/11				none	none	Untagged	none	none			
ethernet1/12				none	none	Untagged	none	none			
ethernet1/13				none	none	Untagged	none	none			
ethernet1/14				none	none	Untagged	none	none			
ethernet1/15				none	none	Untagged	none	none			
ethernet1/16				none	none	Untagged	none	none			
ethernet1/17				none	none	Untagged	none	none			

admin | Logout | Last Login Time: 07/19/2025 15:04:35 | Session Expires Time: 08/18/2025 15:08:55

PA-VM

Ethernet Interface

Interface Name: ethernet1/1

Comment:

Interface Type: Layer3

Netflow Profile: None

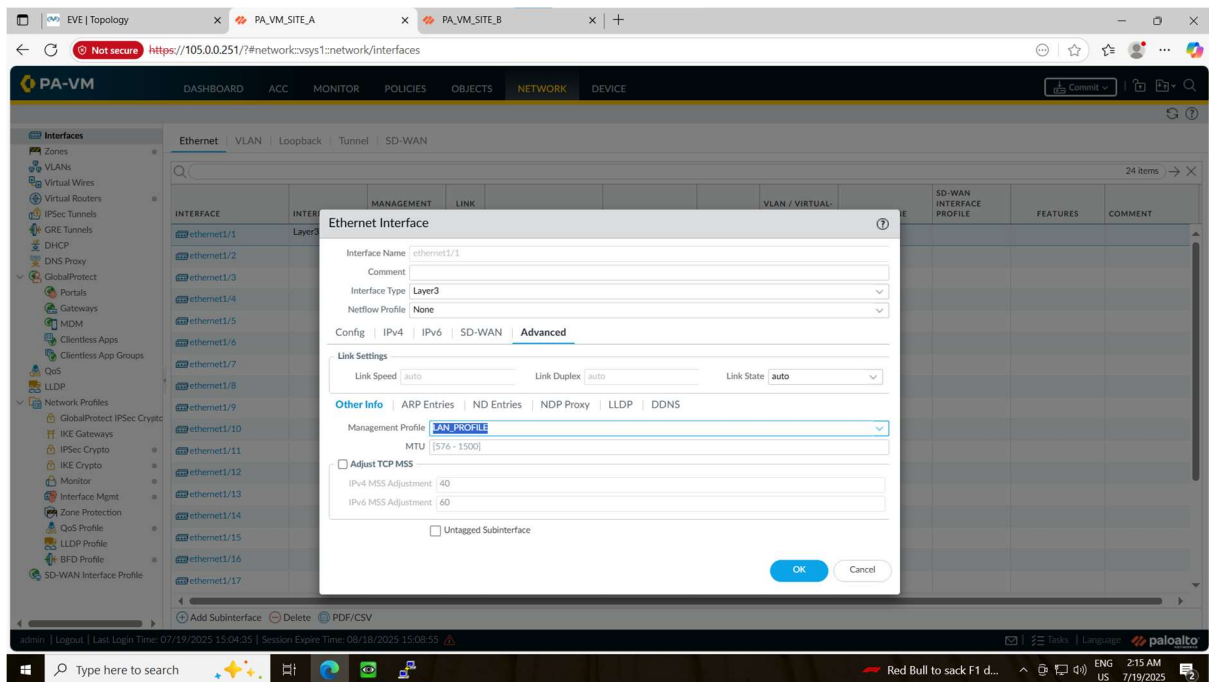
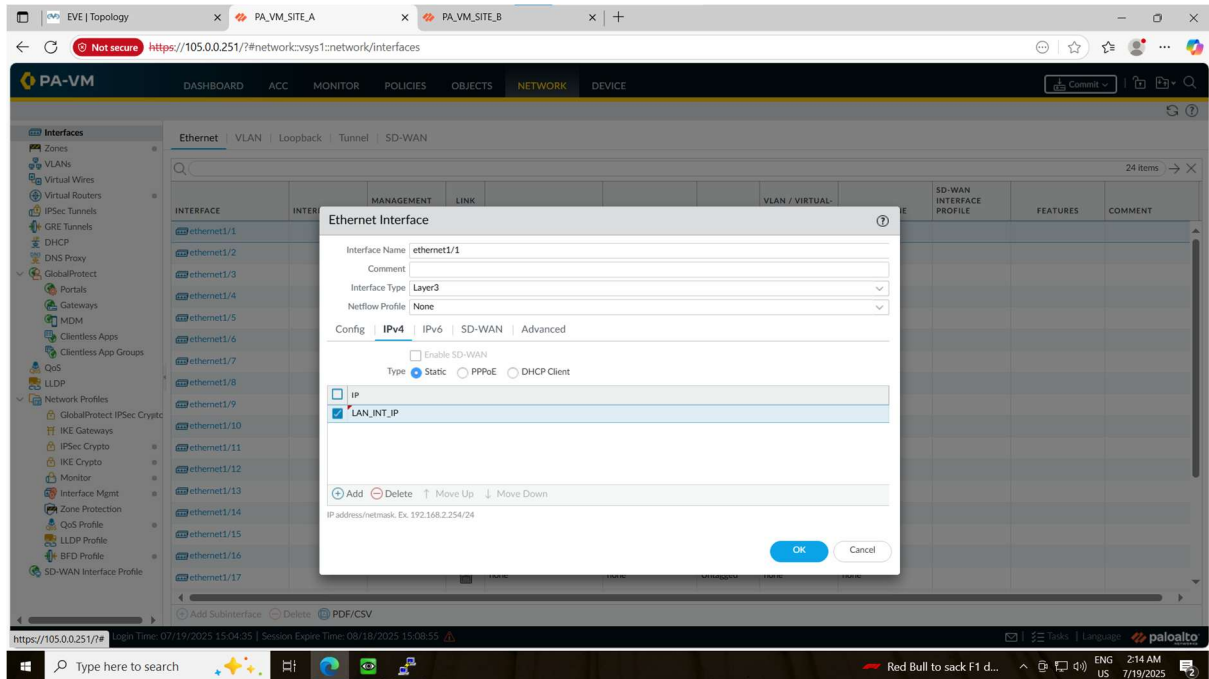
Config | IPv4 | IPv6 | SD-WAN | Advanced

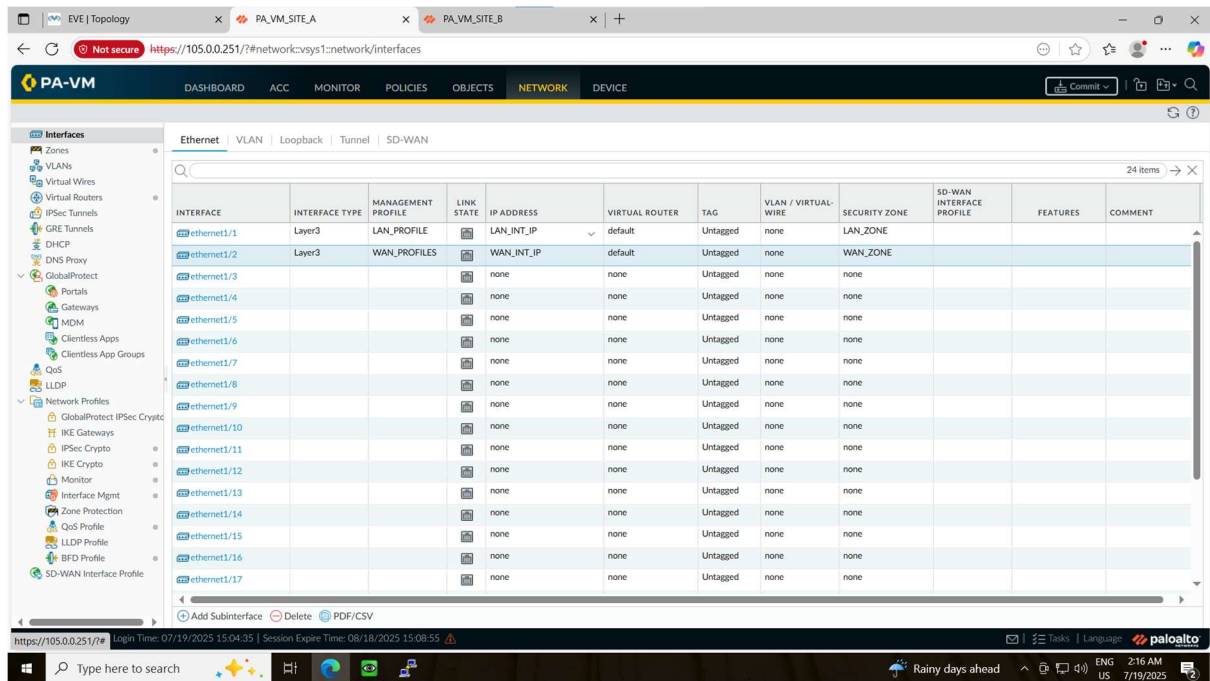
Assign Interface To

Virtual Router: default

Security Zone: LAN_ZONE

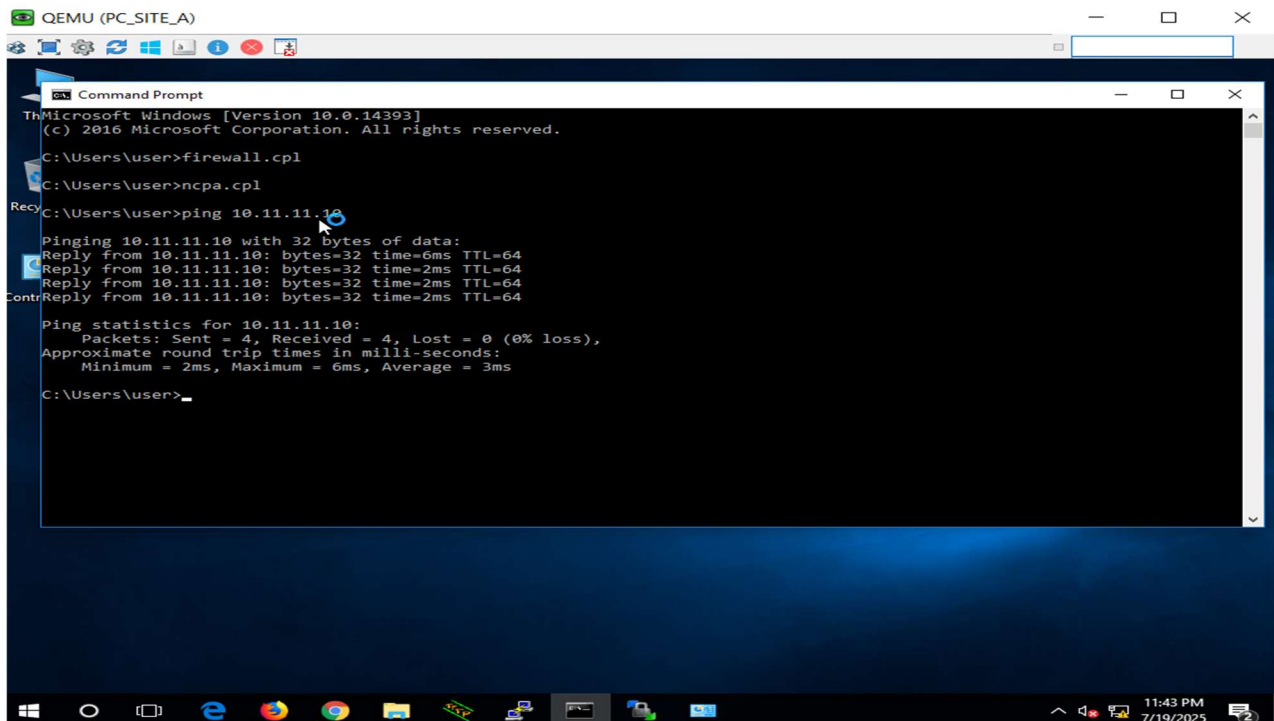
OK Cancel





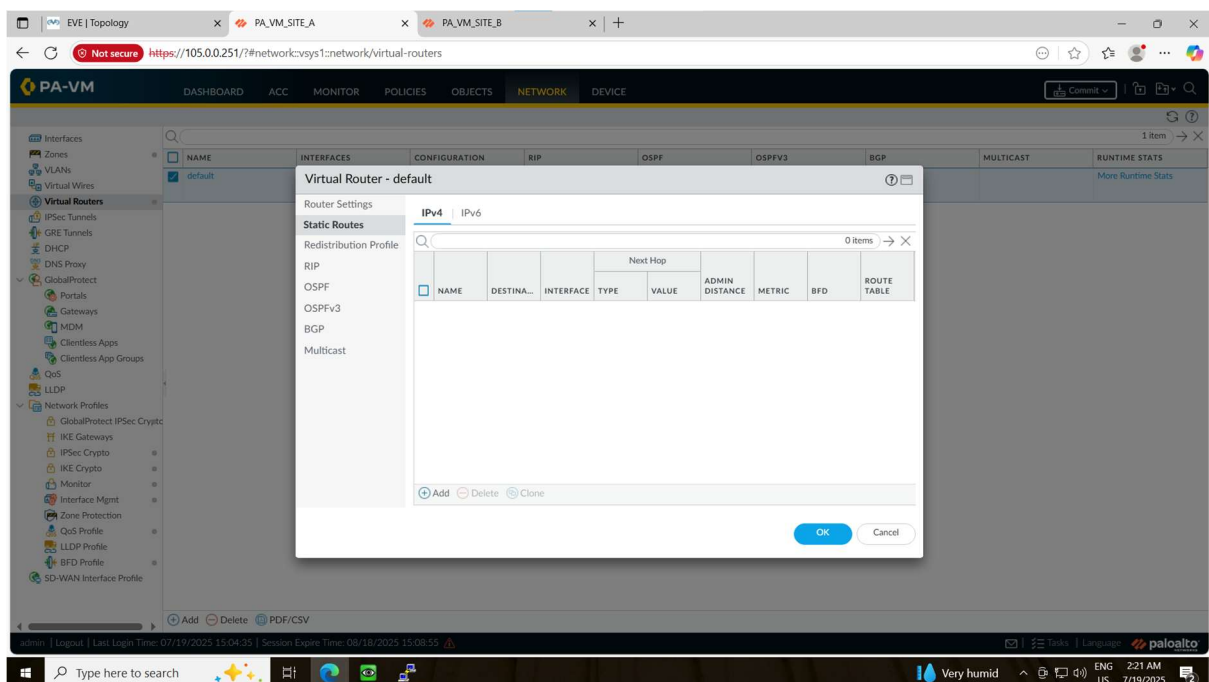
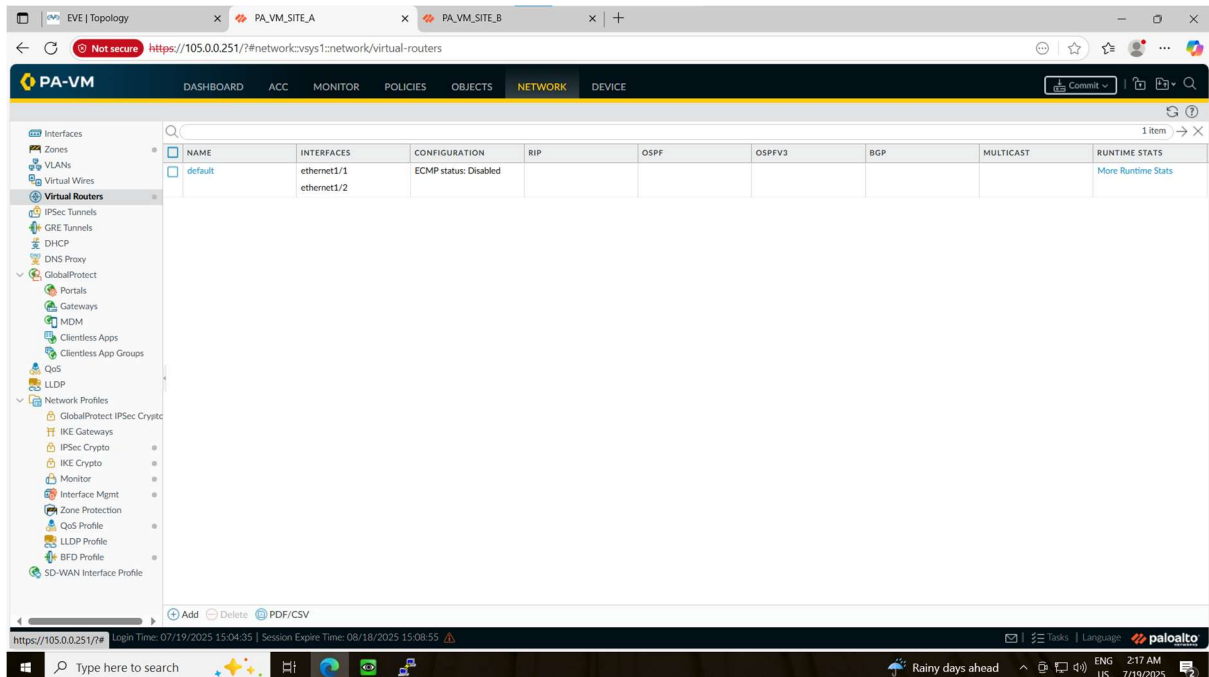
Apply commit for saving all changes

Check reachability from pc to firewall interface in same subnet



6. For public reachability from site a to b create default routing

Network>virtual routers>add



The screenshot shows the Palo Alto VM (PA-VM) web interface. The 'Virtual Router - Static Route - IPv4' dialog is open, displaying the configuration for a static route named 'WAN_ISP_ROUTE'. The destination is 0.0.0/0, the interface is 'None', and the next hop is 1.1.1.2. The admin distance is 10, and the metric is 10. The route table is set to 'Unicast'. The 'Path Monitoring' section is expanded, showing a table with columns for NAME, ENABLE, SOURCE IP, DESTINATION IP, PING INTERVAL (SEC), and PING COUNT. The 'Add' button is visible at the bottom of the table.

Check reachability from site A to site b

```
admin@PA_VM_SITE_A# run ping source 1.1.1.10 host 2.2.2.10
PING 2.2.2.10 (2.2.2.10) from 1.1.1.10 : 56(84) bytes of data.
64 bytes from 2.2.2.10: icmp_seq=1 ttl=63 time=31.9 ms
64 bytes from 2.2.2.10: icmp_seq=2 ttl=63 time=5.83 ms
64 bytes from 2.2.2.10: icmp_seq=3 ttl=63 time=7.28 ms
64 bytes from 2.2.2.10: icmp_seq=4 ttl=63 time=4.87 ms
64 bytes from 2.2.2.10: icmp_seq=5 ttl=63 time=4.95 ms
64 bytes from 2.2.2.10: icmp_seq=6 ttl=63 time=5.67 ms
^C
--- 2.2.2.10 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5005ms
rtt min/avg/max/mdev = 4.872/10.092/31.933/9.799 ms
[edit]
admin@PA_VM_SITE_A#
```

7. VPN Configurations (Phase 1)

Step 1:

Network>network profile>IKE Crypto>

The screenshot shows the Palo Alto VM configuration interface. The left sidebar contains a tree view with categories like Interfaces, Zones, VLANs, Virtual Wires, Virtual Routers, IPsec Tunnels, GRE Tunnels, DHCP, DNS Proxy, GlobalProtect, Portals, Gateways, MDM, Clientless Apps, Clientless App Groups, QoS, LLDP, Network Profiles, GlobalProtect IPsec Crypto, IKE Gateways, IPsec Crypto, IKE Crypto (selected), Monitor, Interface Mgmt, Zone Protection, QoS Profile, LLDP Profile, BFD Profile, and SD-WAN Interface Profile. The main pane displays a table of IKE Crypto configurations.

NAME	ENCRYPTION	AUTHENTICATION	DH GROUP	KEY LIFETIME
default	aes-128-cbc, 3des	sha1	group2	8 hours
Suite-B-GCM-128	aes-128-cbc	sha256	group19	8 hours
Suite-B-GCM-256	aes-256-cbc	sha384	group20	8 hours

At the bottom of the table, there are buttons: Add, Delete, Clone, and PDF/CSV. The status bar at the bottom shows the login time as 07/19/2025 15:04:35 and session expire time as 08/18/2025 15:08:55.

The screenshot shows the Palo Alto VM configuration interface with the IKE Crypto Profile configuration dialog open. The dialog has a title bar "IKE Crypto Profile" and a close button. It contains two main sections: "DH GROUP" and "AUTHENTICATION".

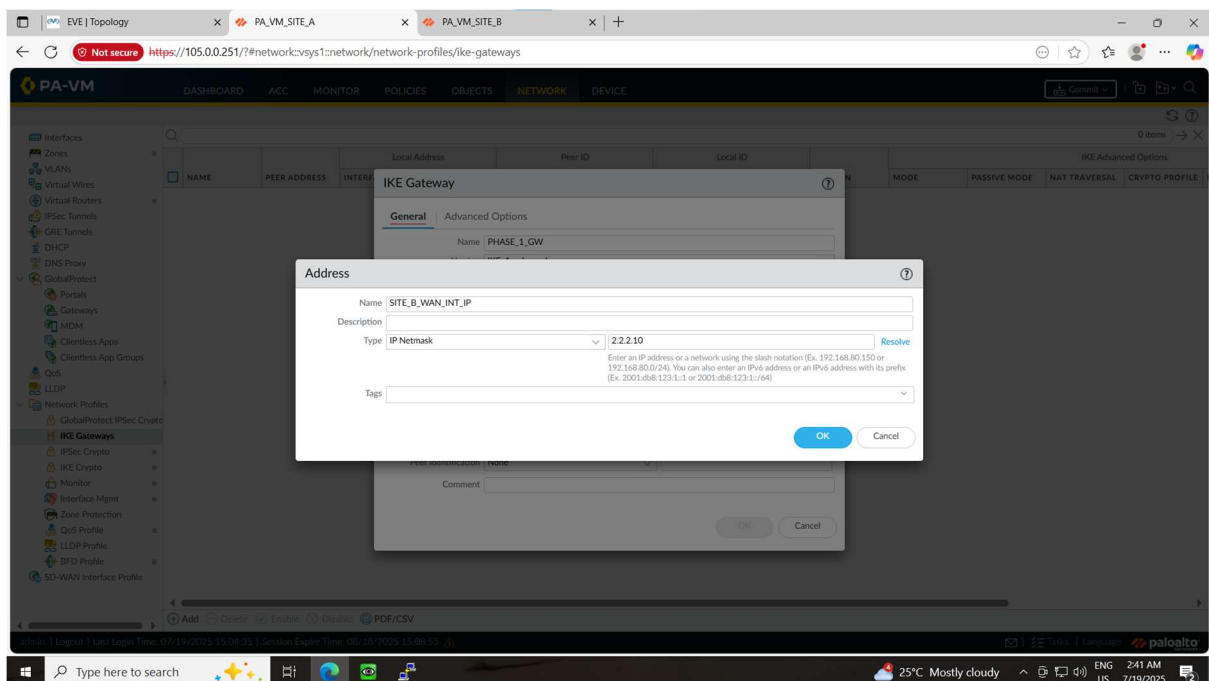
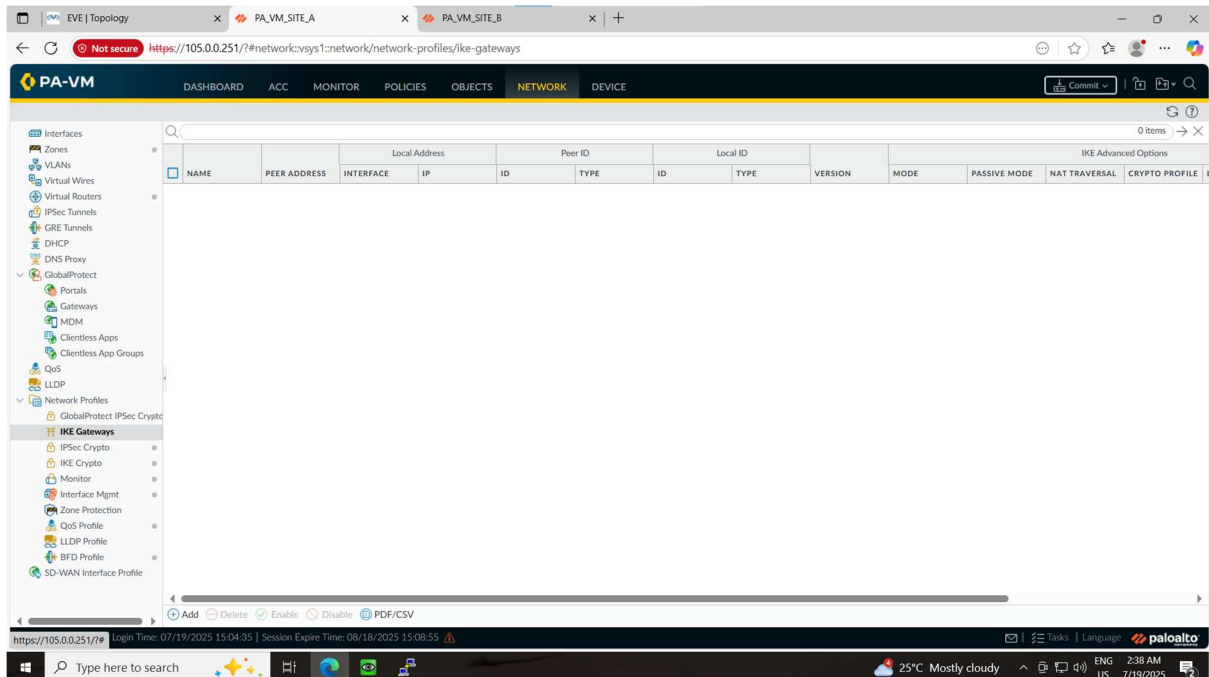
DH GROUP: A list box containing "group5" and "group14". "group14" is selected. Below the list box are buttons: Add, Delete, Move Up, and Move Down.

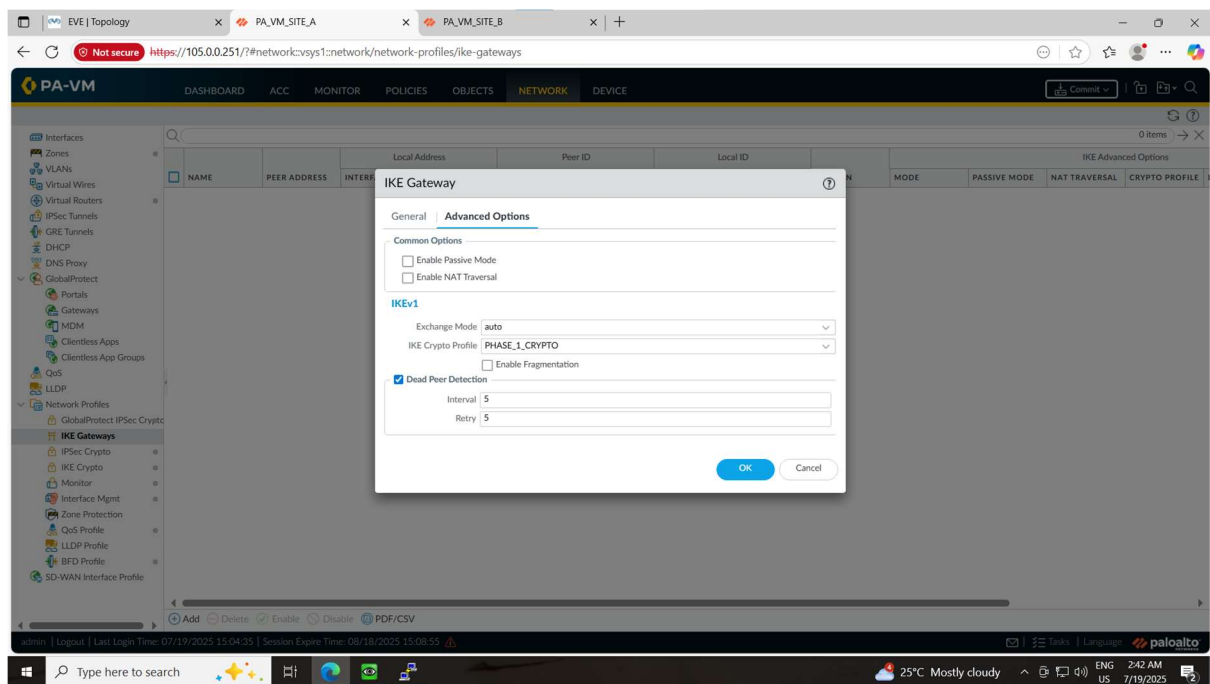
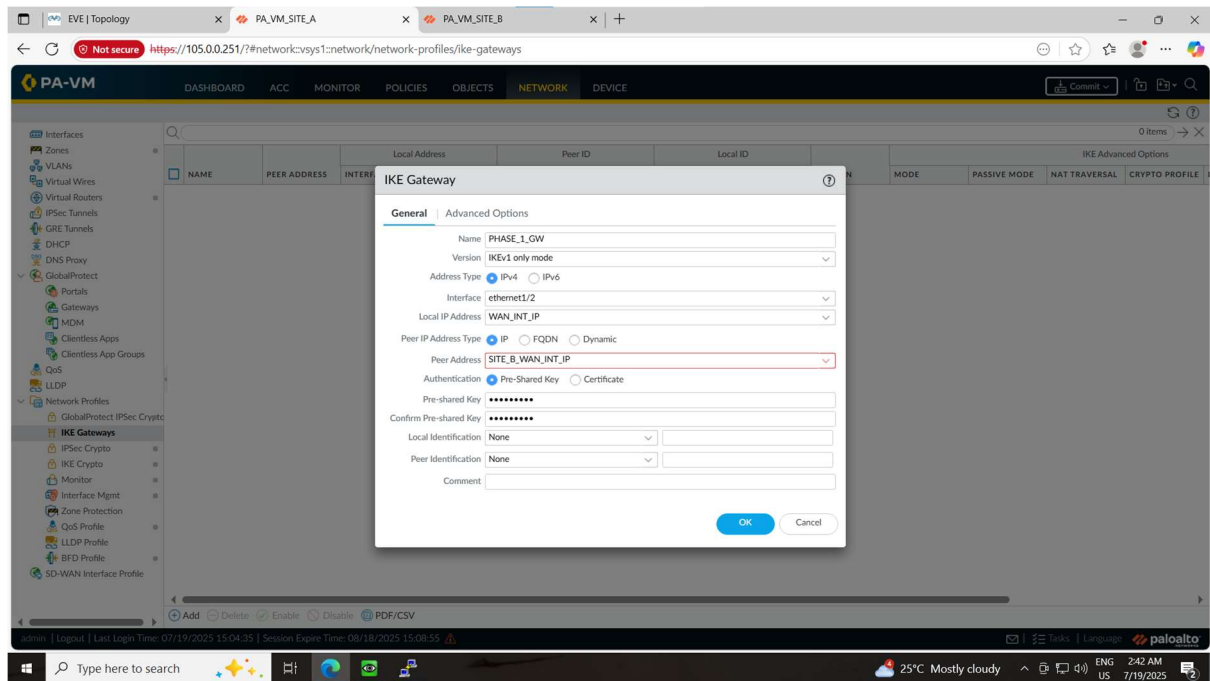
AUTHENTICATION: A list box containing "sha1" and "sha256". "sha256" is selected. Below the list box are buttons: Add, Delete, Move Up, and Move Down.

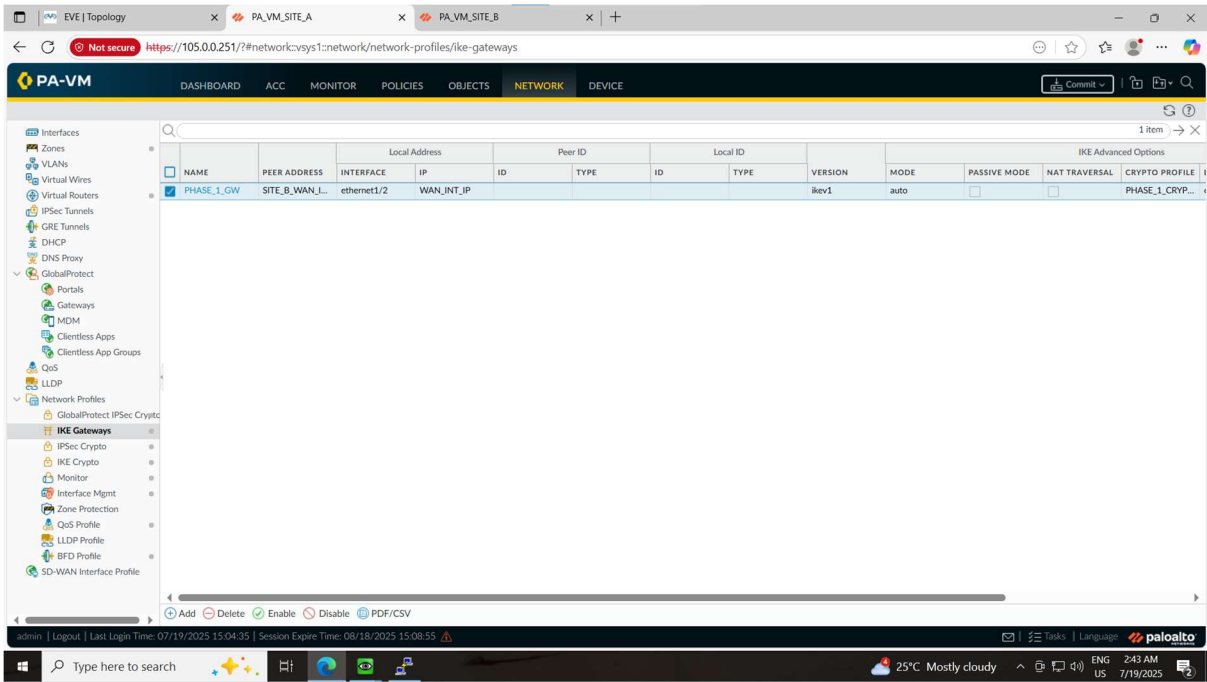
ENCRYPTION: A list box containing "aes-128-cbc" and "aes-192-cbc". "aes-192-cbc" is selected. Below the list box are buttons: Add, Delete, Move Up, and Move Down.

Timers: A section with a "Key Lifetime" dropdown set to "Hours" and a value of "24". Below it is a "Minimum lifetime" field set to "3 mins". There is also an "IKEv2 Authentication" dropdown set to "Multiple".

At the bottom right of the dialog are "OK" and "Cancel" buttons.

Step 2:**Network>network profile>IKE Gateways**





8. VPN Configurations (Phase 2)

Step 1:

Network>network>profile>IPsec Crypto

The screenshot shows the Palo Alto VM web interface. The left sidebar displays a tree view of configuration categories, with 'Network Profiles' expanded and 'IPsec Crypto' selected. The main panel shows a table of IPsec Crypto profiles. The table has columns: NAME, ESP/AH, ENCRYPTION, AUTHENTICATION, DH GROUP, LIFETIME, and LIFESIZE. Three profiles are listed: 'default', 'Suite-B-GCM-128', and 'Suite-B-GCM-256'. The 'default' profile is selected.

NAME	ESP/AH	ENCRYPTION	AUTHENTICATION	DH GROUP	LIFETIME	LIFESIZE
default	ESP	aes-128-cbc, 3des	sha1	group2	1 hours	
Suite-B-GCM-128	ESP	aes-128-gcm	none	group19	1 hours	
Suite-B-GCM-256	ESP	aes-256-gcm	none	group20	1 hours	

The screenshot shows the 'IPsec Crypto Profile' configuration dialog box. The 'Name' field is set to 'PHASE_2_CRYPTQ'. The 'IPsec Protocol' is set to 'ESP'. The 'ENCRYPTION' section is expanded, showing 'aes-128-cbc' selected. The 'AUTHENTICATION' section is expanded, showing 'sha1' selected. The 'DH Group' is set to 'group5'. The 'LifETIME' is set to '1' hour. The 'LIFESIZE' is set to '1' MB. The 'Enable' checkbox is checked. The 'OK' button is highlighted.

EVE | TopologyPA_VM_SITE_APA_VM_SITE_B

Not securehttps://105.0.0.251/?#network:vsys1:network/network-profiles/ipsec-crypto

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Comment

4 Items

InterfacesZonesVLANsVirtual WiresVirtual RoutersIPSec TunnelsGRE TunnelsDHCPDNS ProxyGlobalProtectPortalsGatewaysMDMClientless AppsClientless App GroupsQoSLLDPNetwork ProfilesGlobalProtect IPSec CryptoIKE GatewaysIPSec CryptoIKE CryptoMonitorInterface MgmtZone ProtectionQoS ProfileLLDP ProfileBFD ProfileSD-WAN Interface Profile

NAME	ESP/AH	ENCRYPTION	AUTHENTICATION	DH GROUP	LIFETIME	LIFESIZE
default	ESP	aes-128-cbc, 3des	sha1	group2	1 hours	
Suite-B-GCM-128	ESP	aes-128-gcm	none	group19	1 hours	
Suite-B-GCM-256	ESP	aes-256-gcm	none	group20	1 hours	
PHASE_2_CRYPT0	ESP	aes-128-cbc	sha1	group5	1 hours	

AddDeleteClonePDF/CSV

admin | Logout | Last Login Time: 07/19/2025 15:04:35 | Session Expire Time: 08/18/2025 15:08:55

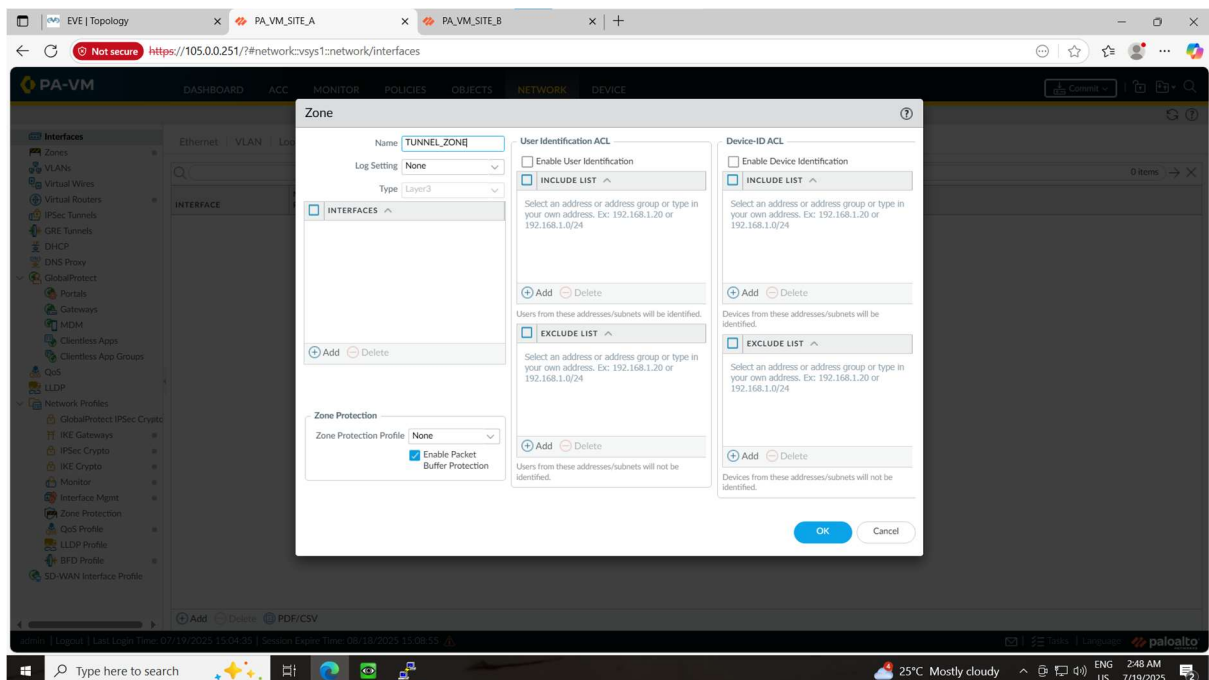
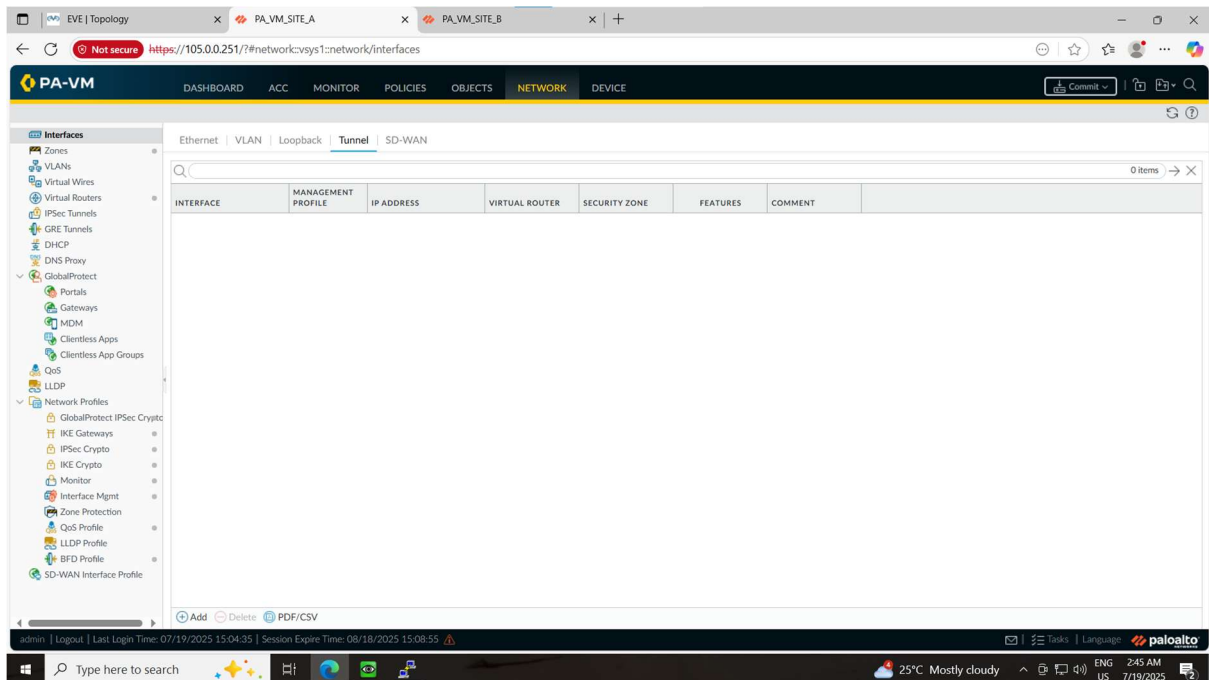
TasksLanguagepalosalto

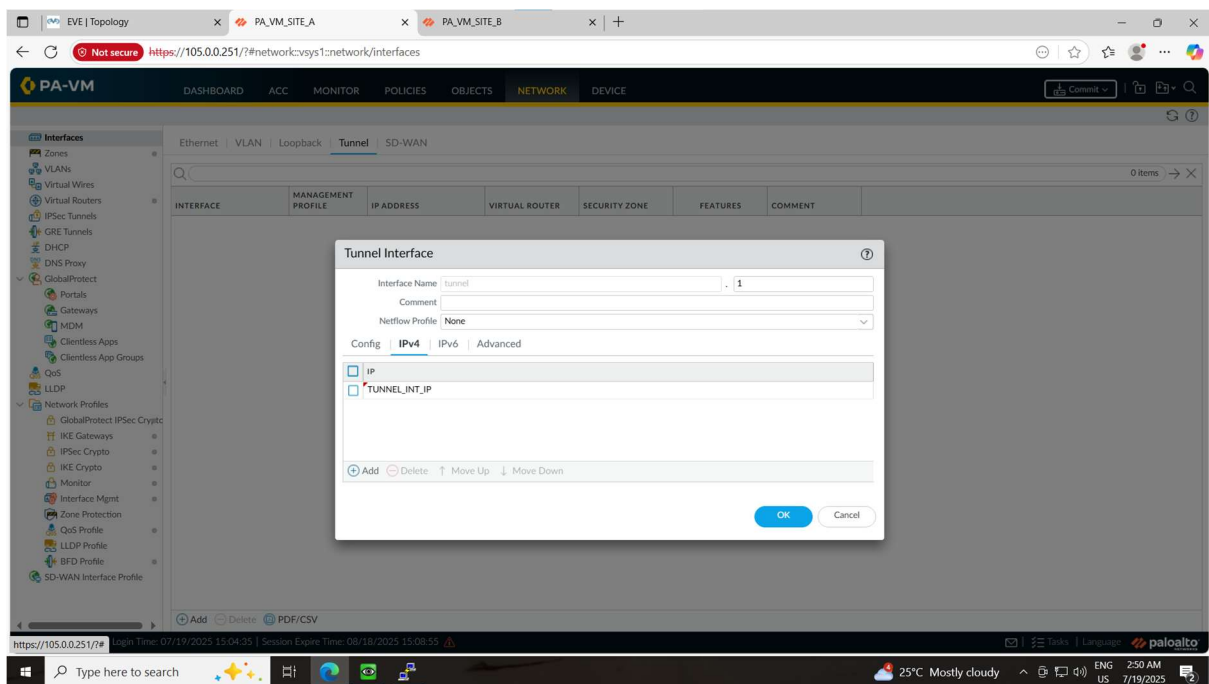
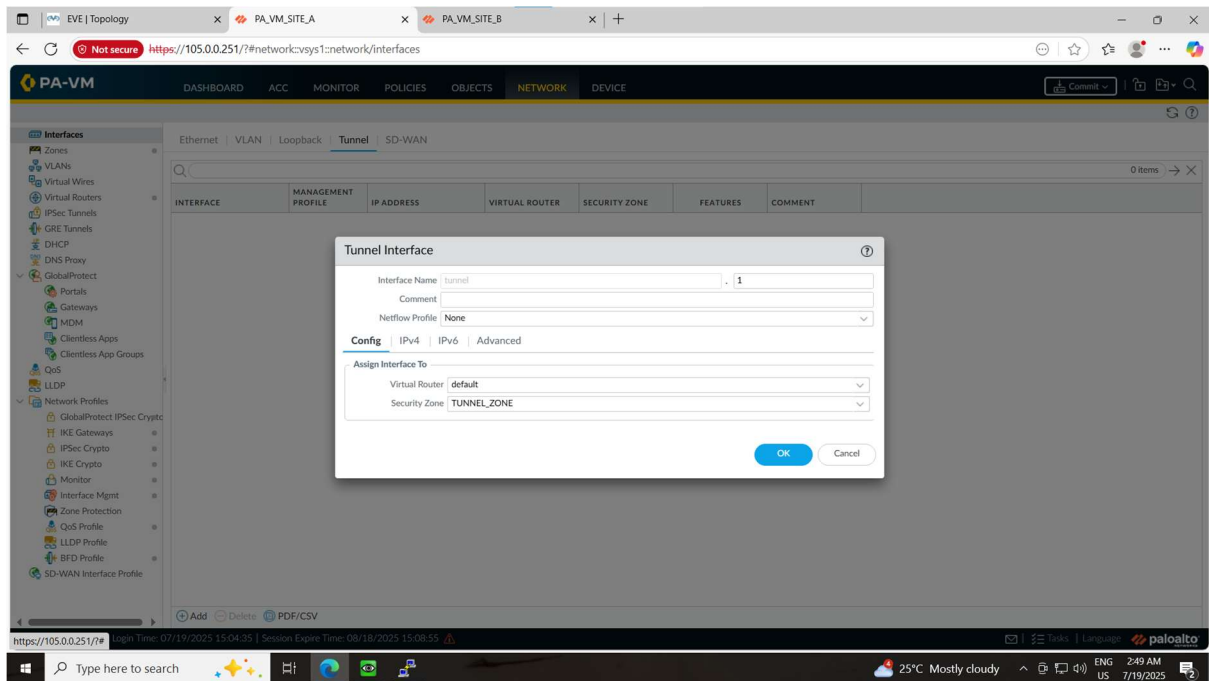
Type here to search25°C Mostly cloudyENG2:45 AMUS7/19/2025

Step:2

To create tunnel

Network>interfaces>tunnel>add





The screenshot displays the Palo Alto Networks PA-VM web interface. The browser address bar shows the URL <https://105.0.0.251/?#network:vsys1:network/interfaces>. The interface has a dark blue header with the PA-VM logo and navigation tabs: DASHBOARD, ACC, MONITOR, POLICIES, OBJECTS, NETWORK, and DEVICE. A left sidebar lists various configuration categories like Zones, VLANs, Virtual Wires, and Network Profiles. The main content area is titled 'Interfaces' and includes sub-tabs for Ethernet, VLAN, Loopback, Tunnel, and SD-WAN. A modal dialog titled 'Tunnel Interface' is open, showing configuration fields for 'Interface Name' (tunnel), 'Comment', 'Netflow Profile' (None), and 'MTU' (576-1500). The 'Advanced' tab is selected under the 'Config' section. The bottom status bar shows the user is 'admin', the last login time is '07/19/2025 15:04:35', and the session expires at '08/18/2025 15:08:55'. The system clock indicates it is 2:51 AM on 7/19/2025.

PA-VM

DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK DEVICE

Interfaces

Ethernet | VLAN | Loopback | Tunnel | SD-WAN

INTERFACE MANAGEMENT PROFILE IP ADDRESS VIRTUAL ROUTER SECURITY ZONE FEATURES COMMENT

Tunnel Interface

Interface Name tunnel

Comment

Netflow Profile None

Config | IPv4 | IPv6 | Advanced

Other Info

Management Profile TUNNEL_PROFILE

MTU [576 - 1500]

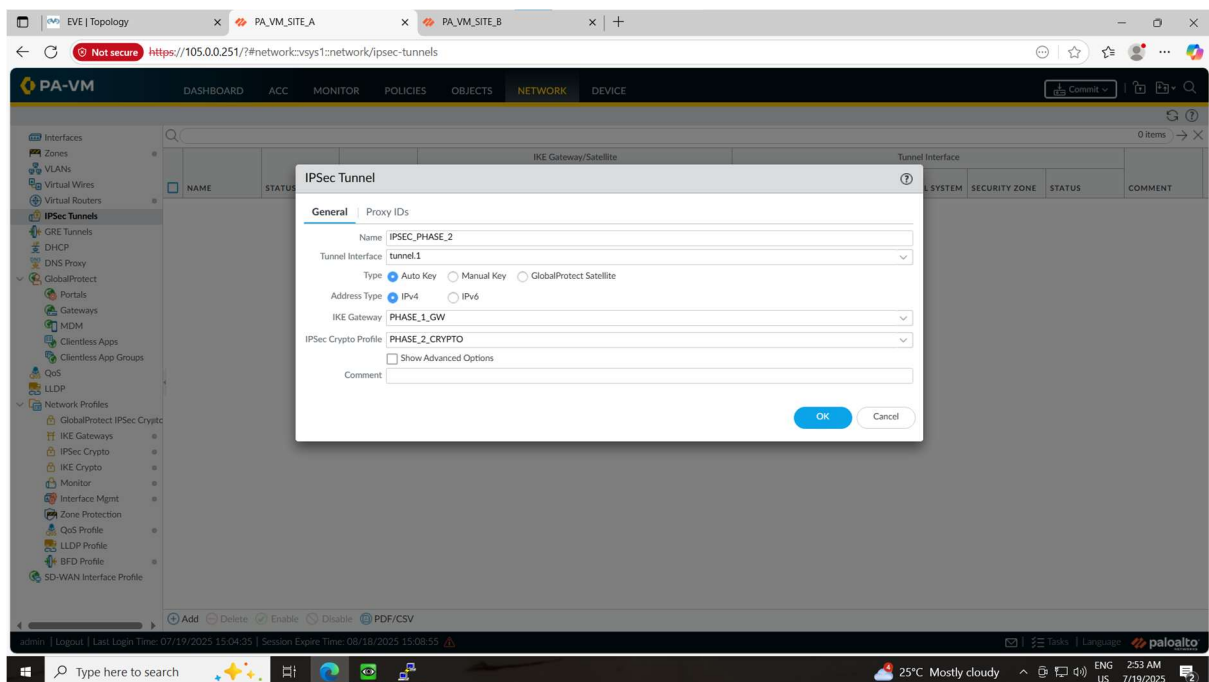
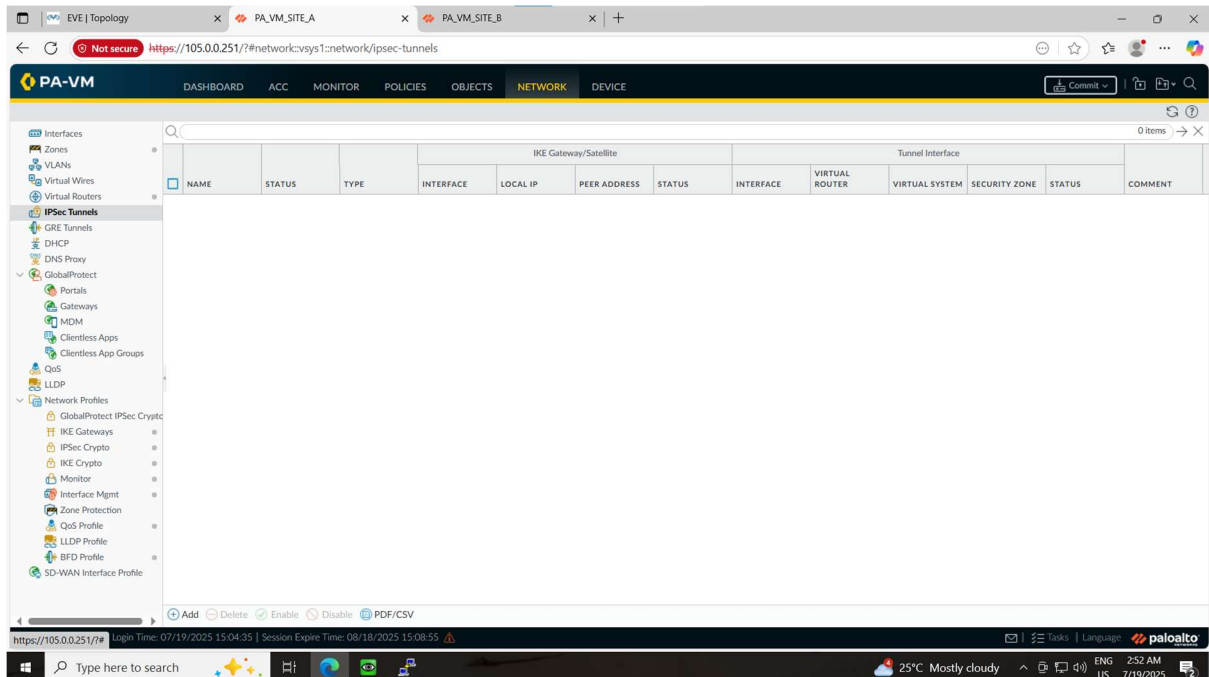
OK Cancel

admin | Logout | Last Login Time: 07/19/2025 15:04:35 | Session Expire Time: 08/18/2025 15:08:55

25°C Mostly cloudy 2:51 AM 7/19/2025

Step 3:

Network>IPsec Tunnels> Add

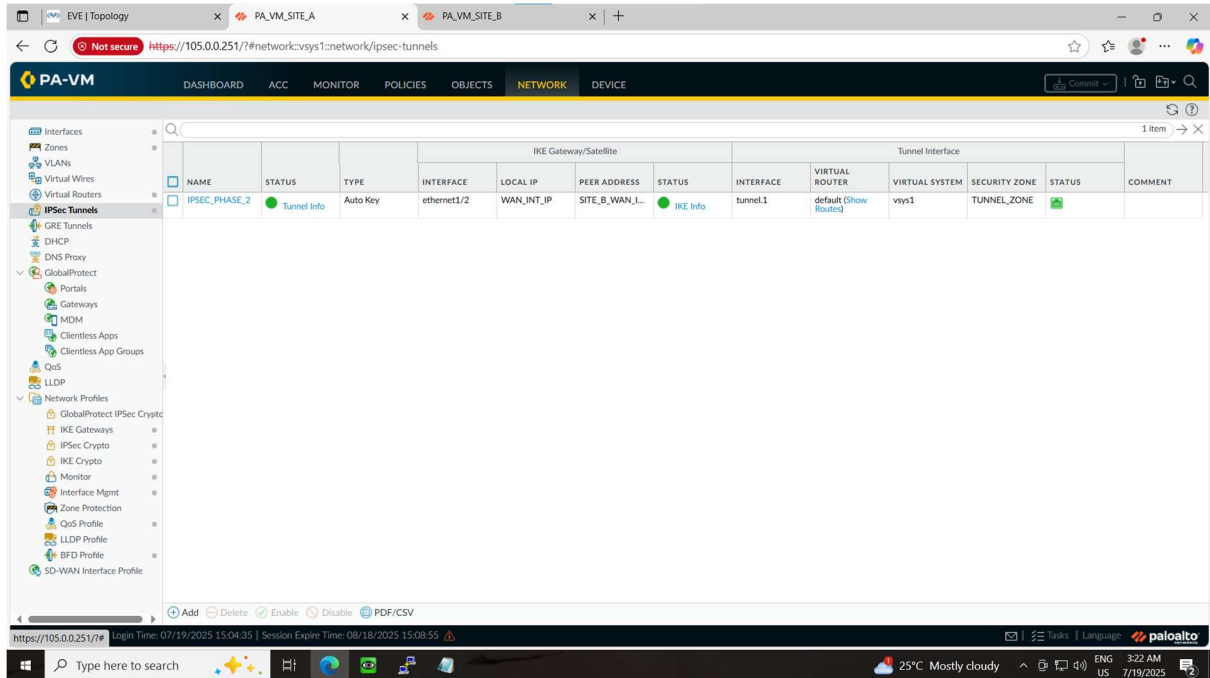


Commit changes and try to ping

```
admin@PA-VM>
admin@PA-VM> ping source 192.168.10.1 host 192.168.10.2
PING 192.168.10.2 (192.168.10.2) from 192.168.10.1 : 56(84) bytes of data.
64 bytes from 192.168.10.2: icmp_seq=2 ttl=64 time=17.7 ns
64 bytes from 192.168.10.2: icmp_seq=3 ttl=64 time=7.71 ns
64 bytes from 192.168.10.2: icmp_seq=4 ttl=64 time=8.89 ns
64 bytes from 192.168.10.2: icmp_seq=5 ttl=64 time=7.18 ns
64 bytes from 192.168.10.2: icmp_seq=6 ttl=64 time=11.0 ns
```

If it can ping, then your tunnel status will turn from red to green

As follows:

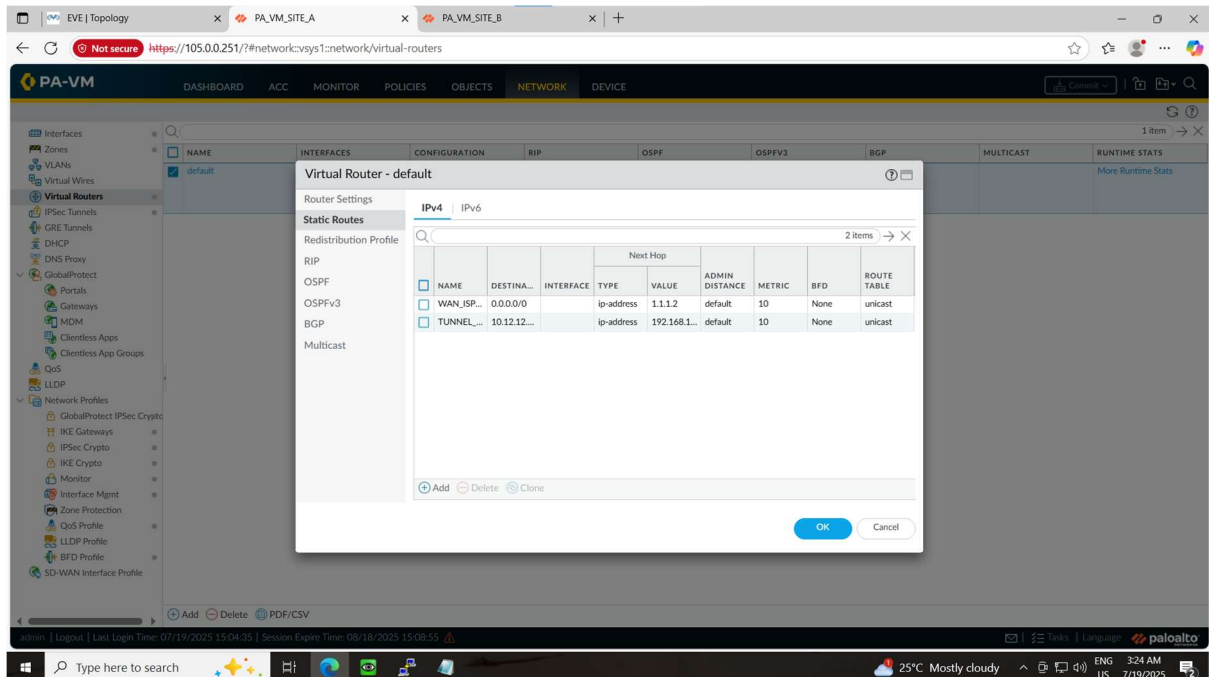


The screenshot displays the Palo Alto Networks PA-VM web interface, specifically the IPsec Tunnels configuration page. The interface is divided into a sidebar on the left and a main content area. The sidebar contains various configuration categories such as Interfaces, Zones, VLANs, Virtual Wires, Virtual Routers, IPsec Tunnels, GRE Tunnels, DHCP, DNS Proxy, GlobalProtect, Portals, Gateways, MDM, Clientless Apps, Clientless App Groups, QoS, LLDP, Network Profiles, GlobalProtect IPsec Crypto, IKE Gateways, IPsec Crypto, IKE Crypto, Monitor, Interface Mgmt, Zone Protection, QoS Profile, LLDP Profile, BFD Profile, and SD-WAN Interface Profile. The main content area shows a table of IPsec Tunnels. The table has columns for NAME, STATUS, TYPE, INTERFACE, LOCAL IP, PEER ADDRESS, STATUS, INTERFACE, VIRTUAL ROUTER, VIRTUAL SYSTEM, SECURITY ZONE, STATUS, and COMMENT. One tunnel, 'IPSEC_PHASE_2', is listed with a green status icon and a 'Tunnel Info' link. The table also includes sub-headers for 'IKE Gateway/Satellite' and 'Tunnel Interface'. The bottom of the interface shows a login time of 07/19/2025 15:04:35 and a session expire time of 08/18/2025 15:08:55. The Palo Alto Networks logo is visible in the bottom right corner.

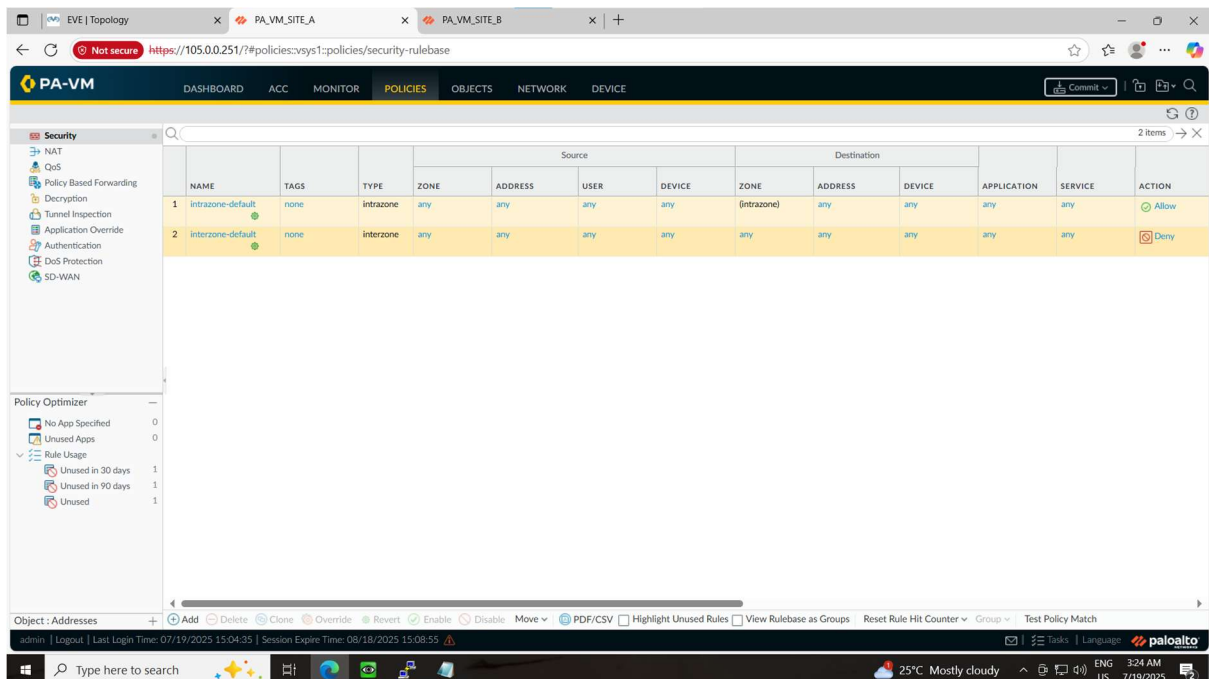
NAME	STATUS	TYPE	INTERFACE	LOCAL IP	PEER ADDRESS	STATUS	INTERFACE	VIRTUAL ROUTER	VIRTUAL SYSTEM	SECURITY ZONE	STATUS	COMMENT
IPSEC_PHASE_2	● Tunnel Info	Auto Key	ethernet1/2	WAN_INT_IP	SITE_B_WAN_I...	● IKE Info	tunnel1	default (Show Routes)	vsys1	TUNNEL_ZONE	●	

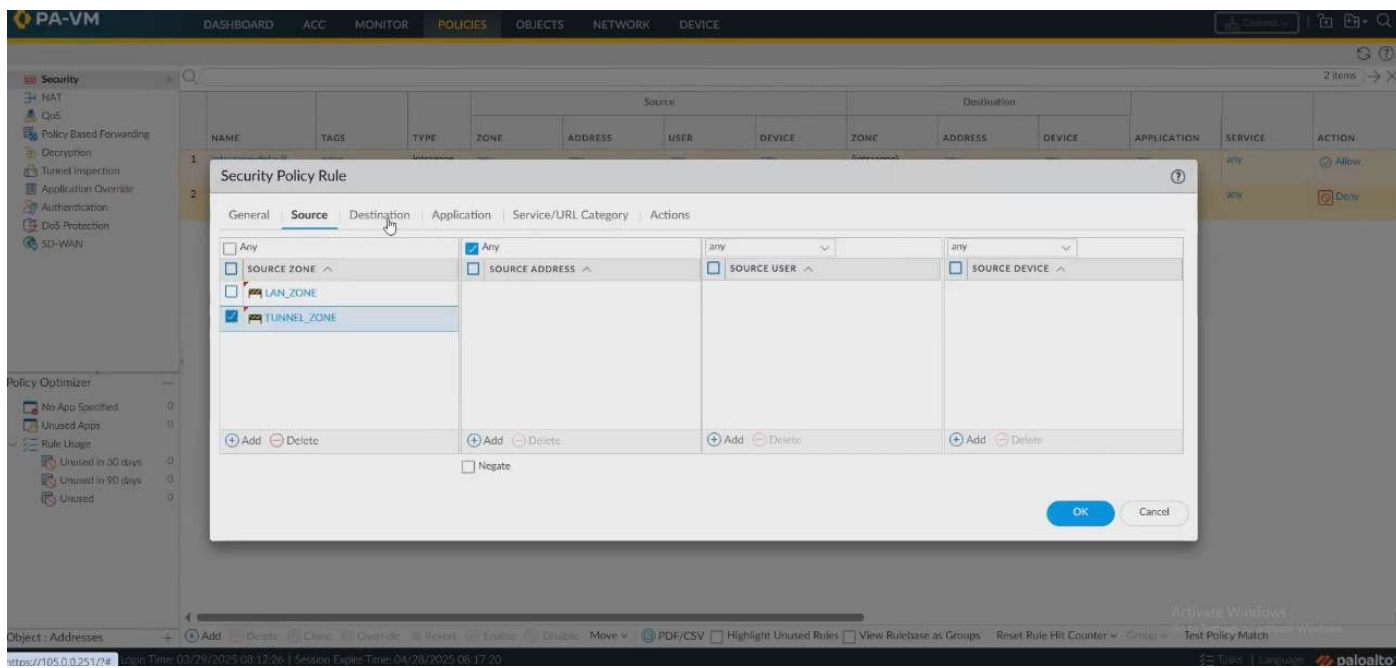
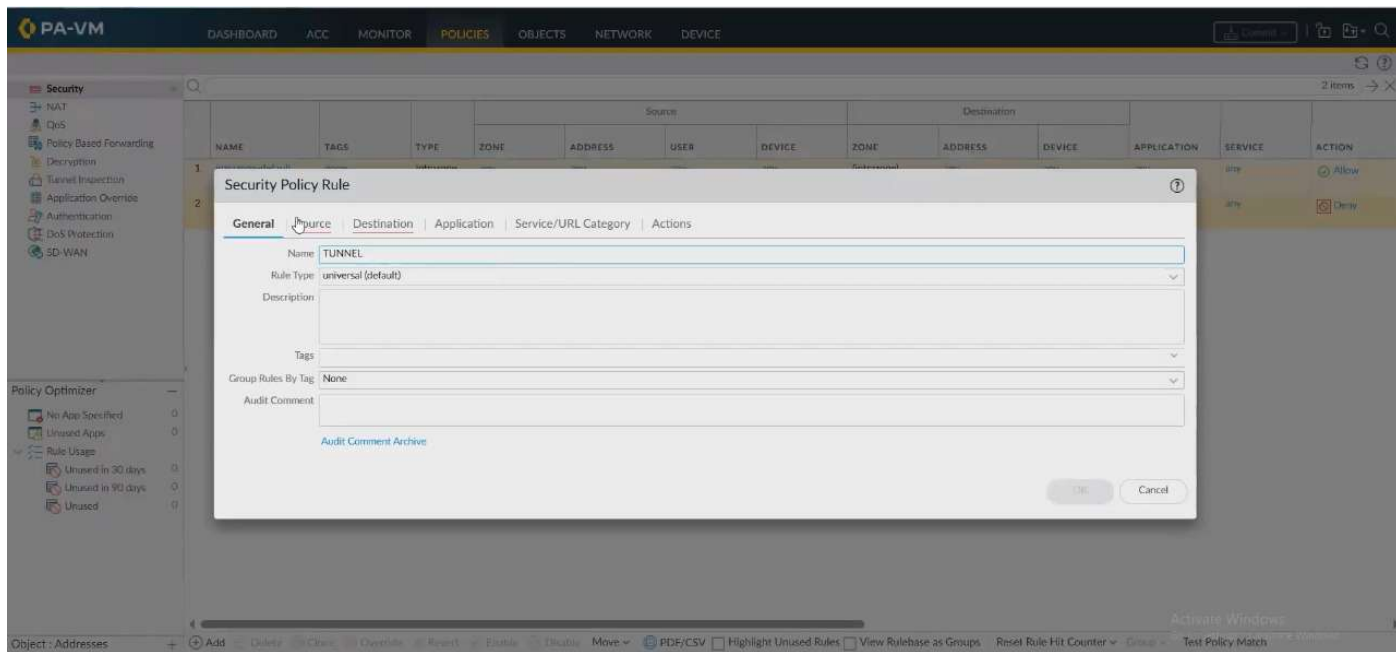
9. Create static routing for site A Lan to site b Lan communication

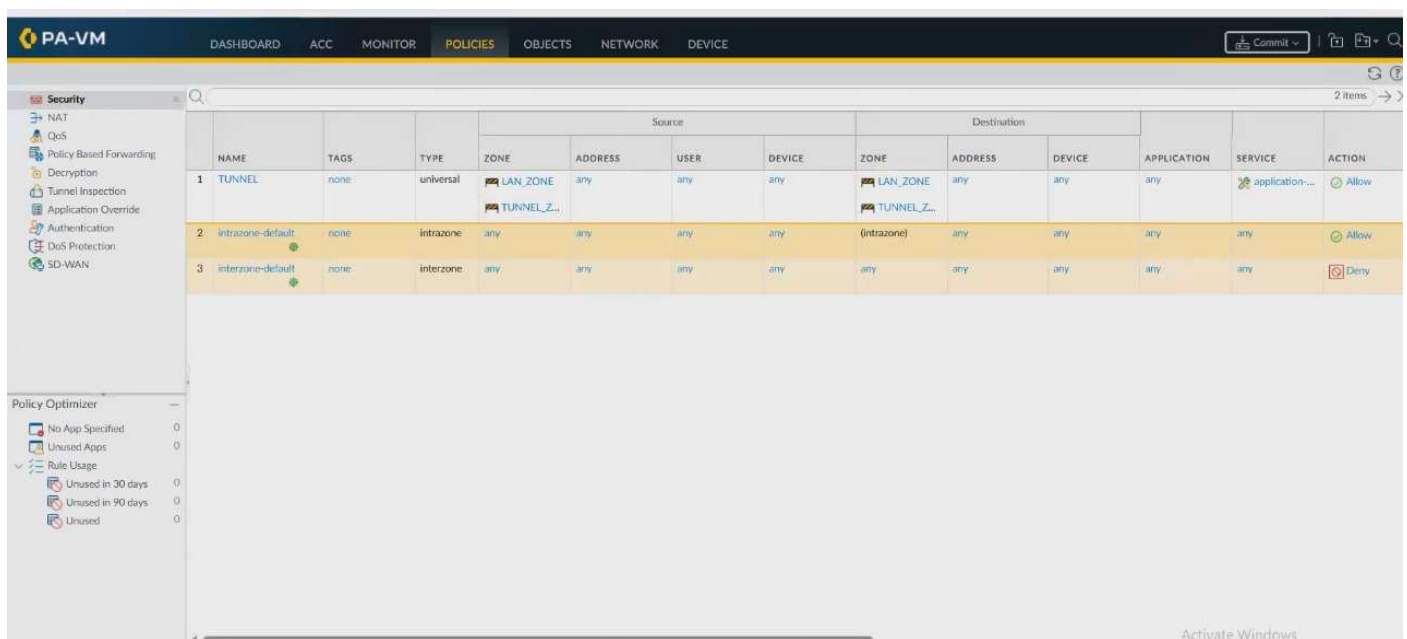
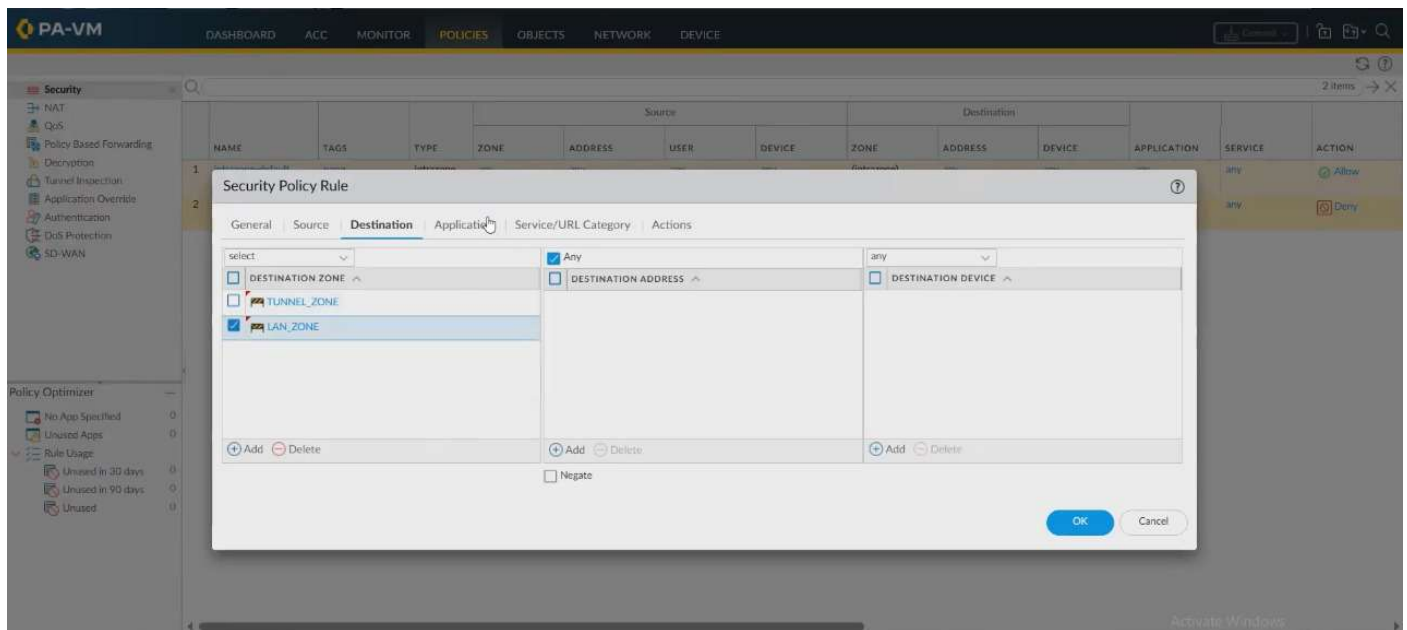
Network> virtual router>default>Static routes> add



10. Create policy for communication







Now commit all changes

REPEAT THE SAME STEPS FROM 1-10 ON PALOALTO VM SITE_B DEVICE

Now you will be able to ping from site A pc to site B pc

